Alaa Hassan Baiomy Wanas

Embedded Software Engineer

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Professional Summary

Highly motivated Embedded Software Engineer with extensive experience in developing and optimizing embedded systems. Proficient in C/C++ programming, real-time operating systems (RTOS), and hardware-software integration. Expertise in microcontroller programming, communication protocols (UART, SPI, I2C, CAN, LIN), and debugging. Strong background in data structures, and algorithms. Passionate about delivering innovative solutions for the automotive and embedded systems industries.

Education

Graduated from Electronics and Communication Engineering **Oct 2021 – Jul 2026**
Al-Azhar University **Cairo, Egypt**

Key Skills

- **Programming Languages:** C, C++, Python, Embedded C, Assembly.
- **Embedded Systems:** RTOS, Device Drivers, Microcontroller Interfacing, Embedded Linux, ROS/ROS2.
- **Development Tools:** Atmel Studio, STM32 CubeIDE, Eclipse, VS Code, Proteus, ROS2 .
- **Debugging & Optimization:** Troubleshootings, Algorithms, Real-Time Debugging, Static Code Analysis.
- **Communication Protocols:** UART, SPI, I2C, Ethernet.
- **Software Development:** Embedded Linux, Unit Testing
- **Version Control:** Git & GitHub.
- **Additional Tools:** MATLAB, JTAG Debugging, Real-Time Trace Tools, Oscilloscopes, firebase.
- **Hardware Proficiency:** ARM Cortex-M, AVR (ATMEGA32), ESP32, Raspberry pi .
- **Soft Skills:** Problem Solving, Team Collaboration, Leadership, Adaptability, Structured Thinking.

Volunteering Experience

- **Member & Technical Support of Embedded System - AZEX** **Oct 2024 – may 2026**
  Al-Azhar University, Cairo, Egypt
  - Implemented communication protocols (UART, SPI, I2C) and applied hardware testing methodologies.
  - Reviewed microcontroller interfacing and developed device drivers.
  - Contributed to the development of a Smart Home project, enhancing skills in embedded system integration.
  - Provided technical support for embedded projects exhibited over 100 student at AZEX.
- **Core Team of Embedded System – Az-SENCS** **Nov 2023 – Sep 2024**
  Al-Azhar University, Cairo, Egypt
  - Learned and applied Embedded C.
  - Assisted in training peers on AVR fundamentals.
- **Instructor & Technical Support Of Robotcs - AZEX** **Oct 2023 – may 2025**
  Al-Azhar University, Cairo, Egypt
  - Delivered lectures Mechanical principles & Arduino programming for robotics to university students, increasing comprehension rates by 95%.
  - Coordinated technical workshops, ensuring the successful participation of over 100 students.

Certifications and Training

- **Embedded Systems Diploma** **Sep 2023 – Mar 2024 Eng. Mahmoud Karem**
  During this diploma studied these topics:
  - Advanced C programming, Data Structures, and Algorithms.
  - RTOS, Device Drivers, ATmega.
- **Advanced Embedded Systems Diploma (In Progress)** **Jul 2024 – Present**

By Eng. Amgad Samir – Senior Embedded Software Engineer at Valeo.

- Studied STM32 microcontroller architecture and ARM Cortex-M core concepts (memory map, processor modes, exceptions, NVIC, etc.).
- Developed drivers for GPIO, UART, SPI, I2C, and DMA using polling, interrupt,and DMA methods.
- Applied advanced debugging techniques, interrupt handling, and driver optimization.

- **Embedded Systems ARM Track "Summer Training"** **Jul 2025 – Aug 2025**

  Information Technology Institute (ITI), Egypt **Hours: 150**

  During this training studied these topics:

  ARM Architecture, Peripherals, Interrupts, Real-Time Concepts, Acoustic Signal, Generation, Communications, IoT, Bootloader, Graduation Project

- **Embedded Linux** **Jul 2025 – Present**

  Currently Studying Embedded Linux, including system architecture, build system architecture, build systems, toolchains, and kernel basics.

  During this diploma studied these topics:

  - System Programming under Linux with Reda Maher.
  - System Architecture, Build Systems, Toolchains, and Kernel Basics with Mahmoud Abdelraouf.

Projects

- **Smart Home Project:** Developed a fully integrated home automation system.
- **Medical Monitor :** ARM Cortex-M4 , Firebase, Flask
- **Defance System :**Radar-integrated launch platform with embedded control (Arduino).
- **Smart Irrigation System:** Automated watering with shift scheduling.
- **Mobile-Controlled Robot:** Bluetooth-enabled Arduino robot with Android app control.
- **traffic Light Controller.**
- **Electronic Voting Machine.**
- **V2X Collision Avoidance System | Graduation Project:** Developed a V2X safety system using STM32, ESP32, and Raspberry Pi running Embedded Linux. Implemented low-latency Inter-Process Communication (IPC) for real-time subsystem synchronization.

LinkedIn Profile

https://www.linkedin.com/in/alaa-hassan-647a35263/

Language

- **Arabic:** Native
- **English:** Very good

Personal Information

- **Date of Birth:** 1/ 1/ 2003
- **Gender: Female**
- **Nationality:** Egyptian